First Midterm Exam

EE241 , Spring 2019

18.3.2019

Q1 (30 pts.)

Simplified block diagram of a load lifting system is given in Figure 1. The weight of load is measured by a 6-bit weight sensor which represents the amount of weight in kg in binary form. The load is lifted with a lifting unit which is powered by Motor-1 and Motor-2. Motor-1 has an internal temperature sensor for protection purposes. It reads the temperature value in Celsius degrees and converts it to binary form.

When the Lift Button is pressed, lifting control unit activates either Motor-1 or Motor-2 or both by giving logic-1 depending on the amount of weight and the temperature value read from Motor-1.

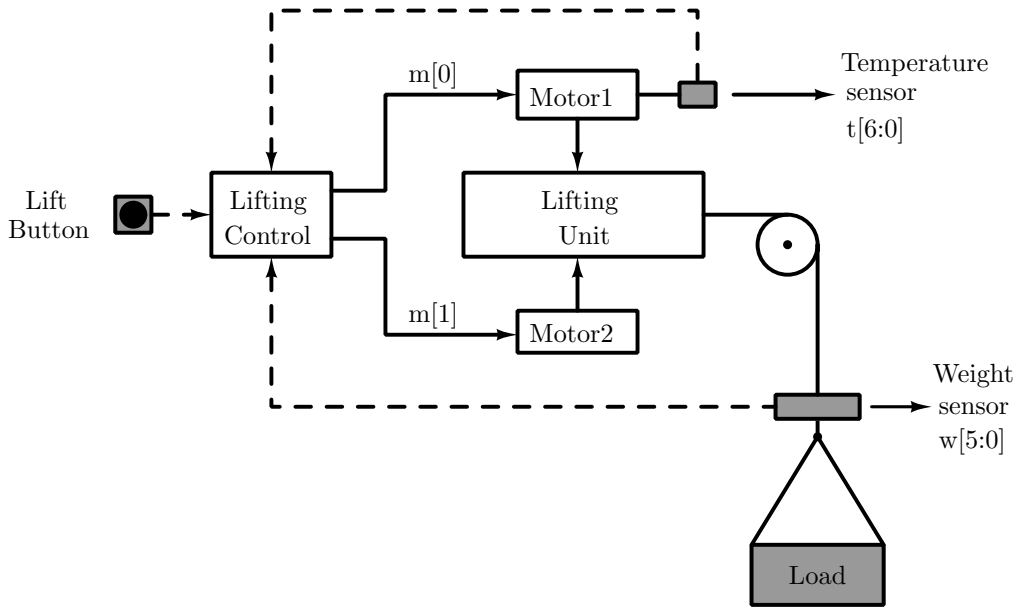


Figure 1: Load lifting system

Design the Lifting Control Unit with following specifications:

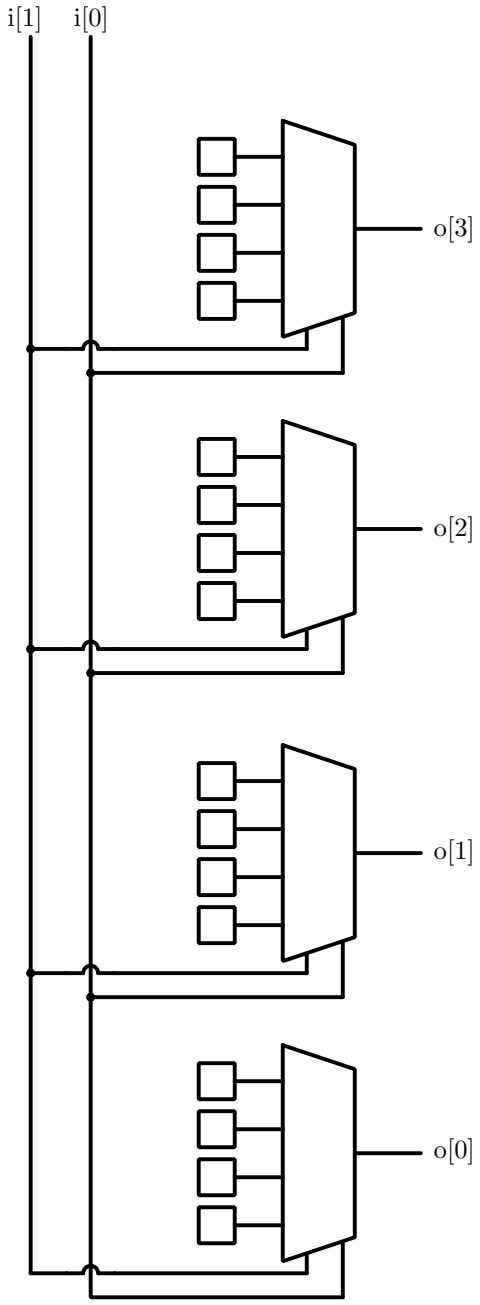
- Motor-1 and Motor-2 will stay in inactive mode, if the Lift Button is not pressed.
- If the button is pressed and the load to be lifted is less than 16 kg, only Motor-1 will be activated.
- If the button is pressed and the load to be lifted is greater than 15 kg, both Motor-1 Motor-2 will be activated.
- When the button is pressed, if the temperature of Motor-1 is greater than 63 ^oC, the load is going to be lifted with Motor-2 only, regardless of the value of the weight.

Write the Verilog description of the problem using dataflow modeling.

Q2 (30 pts.)

a)

Design a logic circuit such that, it will take a 2-bit number i[1:0], multiply it by 5, and display the result on 4-bit number o[3:0]. The design will include four distinct 4 input multiplexers only. Create a truth table and fill in the proper inputs for the multiplexers. No Verilog description is required for the problem.



b) Bonus (20 pts.)

Find a minimum-gate implementation for the problem above. Use minimum number of gates (or connections). Write the Verilog description of the problem.

Q3 (40 pts.)

In cryptography, Caesar's cipher is one of the simplest and most widely known encryption technique. It is a type of substitution cipher in which each letter in the plain text is replaced by a letter some fixed number of positions down the alphabet. You can see an example transform using Caesar cipher in Figure 2.

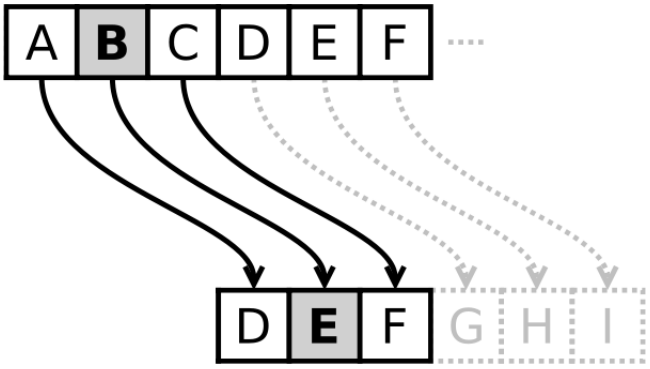


Figure 2: Ciphering example

Write the Verilog description of special Caesar's cipher with following modifications:

- 8-bit input data will contain a 1-bit sign data (MSB) and 7-bit ASCII code.
- Output letter is equal to the input letter shifted by 3. This is done by incrementing its ASCII value by 3.
- Improve your module such that the amount of shift is controlled by the user. The maximum amount of shift is 15.
- Increase the complexity of encryption by checking whether the ASCII code is even or odd. If the ASCII code is even, output letter is equal to the input letter shifted by the value controlled by the user. If the ASCII code is odd, output letter is equal to the ones complement of the input.

		LSB															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
	0	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI
	1	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
	2		!	"	#	$	%	&	'	(	)	*	+	,	-	.	/
M	3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
S	4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
B	5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
	6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
	7	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	DEL

Figure 3: ASCII table